Order-Ten Harmonic-Tail Reconstruction and Polylogarithmic Stieltjes Generators

A one-prime completion theorem, ordinary polygamma identities, and bivariate generating functions for integer-shift Stieltjes correlations

Research continuation prepared for Vladimir Reshetnikov

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Abstract

We complete the fixed smaller-order-ten case of the optimal-prefix harmonic-tail reconstruction problem in the ProveIt polylogarithm program. For every integer $b \geq 6$, the exact rational coefficient matrix $M_{5,b}$ attached to the pair $(p, q) = (10, 2b)$ is nonsingular. The existing eventual theorem already covers $b \geq 1920$. The new input is a single-prime certificate for all remaining 1914 indices: reduction modulo $P = 1{,}000{,}003$ gives a nonzero determinant for every $6 \leq b \leq 1919$. For $6 \leq b < 10$ the source matrix is used directly; for $10 \leq b \leq 1919$ a Padé–Bernoulli reduction replaces it by an exact 5×5 determinant. Bernoulli residues are generated from their defining recurrence, no floating-point arithmetic enters, and the complete residue vector is supplied with a SHA-256 digest and an independent verifier.

The rank theorem yields an infinite family of ordinary convergent identities reconstructing every even zeta coordinate from products of generalized harmonic tails, equivalently products of high polygamma functions. We give an explicit degree-30 identity for $\zeta(2)$ from the pair $(10, 16)$. A separate analytic theorem expresses the generating function of two Hurwitz tails through depth-two multiple polylogarithms and a Gamma-normalized double Mellin integral; the reconstruction identities become boundary values of explicit polylogarithmic differential operators.

Finally, we develop a bivariate regularized multiple-polylogarithm kernel whose Taylor coefficients generate all products $\gamma_r(n+1)\gamma_s(n+1)$ of generalized Stieltjes constants at integer shifts. It contains as a low-order specialization the exact identity

$$\sum_{n\geq 0}(\gamma - H_n)^2 z^n = \frac{(\gamma + \log(1-z))^2 + \mathrm{Li}_2(z)}{1-z}, \qquad |z| < 1,$$

and an anchored antiderivative evaluated explicitly at $z = 1/2$. Every new finite claim is separated from inherited input, and all computational claims have exact reproducibility artifacts.

Contents

1 Scope, provenance, and statement of contributions **4**

1.1 Main results 4

1.2 Proof-status conventions 5

1.3 Why a finite certificate is a proof here 5

2 Harmonic tails, centered moments, and the source matrix **5**
2.1 Late coordinates . . . 6
2.2 The shifted-Hankel reduction . . . 6
2.3 Eventual nonsingularity . . . 8

3 Complete optimal-prefix reconstruction at smaller order ten **8**
3.1 A good prime and exact Bernoulli reduction . . . 8
3.2 The exhaustive residue vector . . . 9
3.3 The completed fixed-order theorem . . . 10

4 Ordinary convergent reconstruction and polygamma identities **10**
4.1 Centered asymptotic coefficients . . . 10
4.2 The rational extraction rows . . . 11

5 A multiple-polylogarithm generator for two harmonic tails **12**
5.1 Centered weights as one differential operator . . . 13

6 Integer-shift Stieltjes correlations from a regularized depth-two kernel **14**
6.1 Single-index order derivatives . . . 14
6.2 The bivariate regularized kernel . . . 14
6.3 Digamma square and an anchored antiderivative . . . 15

7 An explicit $(10, 16)$ reconstruction of $\zeta(2)$ **16**

8 Audit findings, corrections, and integration consequences **17**
8.1 Research-status updates . . . 17
8.2 Certificate wording . . . 18
8.3 Analytic normalization . . . 18
8.4 Novelty and literature scope . . . 18
8.5 Recommended manuscript placement . . . 18

9 Further research questions and exact next targets **19**

A Certificate specification **20**
A.1 Input range and residue meaning . . . 20
A.2 Canonical generation algorithm . . . 20
A.3 Two replay paths . . . 21
A.4 Sentinel values . . . 21

B Coefficient table for the explicit identity **21**

C Reproducibility and evidential roles **22**
C.1 Exact proof checks . . . 22
C.2 Numerical diagnostics . . . 22
C.3 Complete replay and build . . . 22

C.4 Claim classification 23

1 Scope, provenance, and statement of contributions

This report continues the harmonic-tail reconstruction branch of the ProveIt polylogarithm manuscript. The base repository was inspected at commit [1]

`04bf92d507588a26abadbaee1beb1e7350c8bffb` (11 October 2026).

Three earlier continuations are the direct mathematical inputs. The Bernoulli–Tornheim–digamma report established the shifted-Hankel factorization and the reduction of each fixed smaller order $p = 2a$ to an $a \times a$ determinant. The quartic spectral reconstruction report proved effective eventual nonsingularity for every fixed a and computed the sufficient threshold $B_5 = 1920$. The alternating-tail report developed a five-square order-ten contraction and proved nonsingularity outside one 2-adic progression, but explicitly left that progression and unrestricted order ten open. The quartic report then listed completion of $6 \le b < 1920$ as its first concrete research target [2, 3, 4].

The present report resolves exactly that target. It also records two analytic consequences which are useful independently of the determinant problem. The first is a multiple-polylogarithm generating function for two harmonic tails and its weighted-moment operator calculus. The second is a regularized depth-two multiple-polylogarithm generator for integer-shift Stieltjes correlations. These formulas were not found in the inspected manuscript or continuation packages. No claim of priority over all external literature is made; the novelty claim in this report is deliberately local: the formulas and proofs are new relative to the audited ProveIt corpus, and the exact order-ten completion answers an open problem stated there.

1.1 Main results

The principal statements are the following.

M1. Complete order-ten reconstruction. For every integer $b \ge 6$, the rational coefficient matrix $M_{5,b}$ is nonsingular. Hence the first $q = 2b$ centered moments attached to $(p, q) = (10, 2b)$ have full formal rank q.

M2. One-prime finite certificate. For the finite range $6 \le b \le 1919$, reduction modulo

$$P = 1{,}000{,}003$$

certifies every determinant at once. The 1914 residues are all nonzero. Their canonical comma-separated encoding has digest

`3977e54f5b43db43e8f080c32f2e5b22beac54412cca32ff5d03d84975111836`.

The certificate is independently regenerated from the Bernoulli recurrence and exact modular Gaussian elimination.

M3. Ordinary reconstruction identities. For each $b \ge 6$ and $0 \le k < b$, explicit rational polynomials $P_{b,k}$ and $Q_{b,k}$ satisfy

$$\sum_{n=0}^{\infty} \Big(P_{b,k}(n + \tfrac{1}{2})[\zeta(10) - H_n^{(10)}][\zeta(2b) - H_n^{(2b)}] - Q_{b,k}(n + \tfrac{1}{2}) \Big) = \zeta(2k),$$

where $\zeta(0) = -1/2$. The same identity is a product-polygamma formula after replacing each tail by a polygamma derivative.

M4. Two-tail polylogarithmic generator. For $p, q > 1$ and $|z| < 1$, the ordinary generating function of $\zeta(p, n+1)\zeta(q, n+1)$ is a divided difference of depth-two multiple polylogarithms. Polynomially weighted centered moments are obtained by the single Euler operator $z\,d/dz + 1/2$.

M5. Stieltjes correlation generator. A bivariate pole-subtracted depth-two polylogarithm kernel is holomorphic at $(\alpha, \beta) = (0, 0)$, and its Taylor coefficients generate all products $\gamma_r(n+1)\gamma_s(n+1)$. The theorem includes explicit anchored primitives and special values.

1.2 Proof-status conventions

A statement marked *inherited* is quoted from an existing ProveIt report and used as an input. A statement marked *new* is proved here. A numerical check is never used as a proof gate. The finite certificate operates in an exact prime field; the high-precision check of the explicit $(10, 16)$ identity is only a diagnostic. Formal rank means rank in the rational coordinate space specified by the source construction. It does not assert arithmetic independence of the numbers $\zeta(2k)$ or of any periods obtained after numerical evaluation.

1.3 Why a finite certificate is a proof here

The eventual theorem converts the order-ten conjecture into a finite problem before any computation begins. The retained determinant has fixed size five, every entry is an explicitly displayed rational function of b and finitely many Bernoulli numbers, and all denominators are controlled. Reduction modulo a prime which divides none of those denominators is an exact ring homomorphism. A nonzero modular determinant therefore proves that the original rational determinant is nonzero. The certificate is not evidence extrapolated from samples; it is an exhaustive exact verification of the complete finite set left by the analytic theorem.

2 Harmonic tails, centered moments, and the source matrix

For an integer $s \geq 2$, write

$$A_s(n) = \zeta(s) - H_n^{(s)} = \zeta(s, n+1), \qquad H_n^{(s)} = \sum_{m=1}^{n} \frac{1}{m^s}, \qquad x_n = n + \frac{1}{2}. \tag{1}$$

Thus $A_s(n)$ is simultaneously a generalized harmonic tail, a Hurwitz-zeta value, and a normalized polygamma value [5]:

$$A_s(n) = \frac{(-1)^s}{(s-1)!}\psi^{(s-1)}(n+1). \tag{2}$$

For even s, the sign in (2) is positive.

For even $p, q \geq 2$, the source family is

$$D_{p,q}(w) = \sum_{n=0}^{\infty} A_p(n)A_q(n)x_n^{-w}, \qquad \Re w > 3 - p - q. \tag{3}$$

The previous reports continue this function meromorphically and prove that it is regular at the even nonpositive arguments needed below. We write

$$D_r = D_{p,q}(-2r), \qquad r \geq 0.$$

The centered Bernoulli transform is

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r, \qquad j \geq 0. \tag{4}$$

Its inverse is

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}\left(\frac{1}{2}\right) C_j. \tag{5}$$

Both transforms are triangular with nonzero diagonal.

2.1 Late coordinates

Put $p = 2a$, $q = 2b$, with $1 \leq a < b$, and introduce formal labels

$$Z_0 = \zeta(0) = -\frac{1}{2}, \qquad Z_k = \zeta(2k) \quad (1 \leq k < b).$$

The source calculation proves that the late centered moments have the rational coordinate expansion

$$C_{b+t} = \sum_{k=0}^{b-1} (M_{a,b})_{tk} Z_k, \qquad 0 \leq t < b. \tag{6}$$

Set

$$\mu_j = \begin{cases} B_{2j}/(2j)!, & j \geq 0, \\ 0, & j < 0. \end{cases}$$

Then the exact matrix in the normalization used by the current continuations is

$$(M_{a,b})_{tk} = \mu_{t+k-a+1} \left\{ \frac{(2b+2t-2a+1)!}{(2b-2k)!} + \mathbf{1}_{k<a} \frac{(2t+1)!}{(2a-2k)!} \right\}, \quad 0 \leq t, k < b. \tag{7}$$

Every factorial argument in (7) is nonnegative in the range where its term is present. Negative Bernoulli indices are zero by convention.

The first b coordinates $C_0, \ldots, C_{b-1}$ are free formal labels, while (6) describes the next b. Hence the coordinate matrix of $C_0, \ldots, C_{2b-1}$ is block diagonal with blocks I_b and $M_{a,b}$. Since the transform (4) is triangular, we obtain the fundamental criterion.

Proposition 2.1 (Source rank criterion; inherited)**.** *The first $q = 2b$ centered moments $D_0, \ldots, D_{2b-1}$ have full formal rank $2b$ if and only if*

$$\det M_{a,b} \neq 0.$$

The statement is entirely rational and makes no claim of numerical linear independence among the values assigned to the coordinate labels.

2.2 The shifted-Hankel reduction

The Bernoulli core of (7) is the shifted Hankel matrix

$$H_{a,b} = (\mu_{t+k-a+1})_{0 \leq t,k < b}. \tag{8}$$

The source factorization has the form

$$M_{a,b} = R(H_{a,b} + U)E,$$

where R and E are nonsingular diagonal matrices and U has rank at most a. Orthogonal-polynomial row operations based on the Padé approximants to

$$\frac{\sqrt{z}}{2}\coth\frac{\sqrt{z}}{2}=\sum_{j\ge 0}\mu_j z^j$$

annihilate the last $b-a$ columns in the final a transformed rows. Consequently, the full determinant reduces to a retained determinant of size a. The reduction also proves that the unperturbed matrix $H_{a,b}$ is nonsingular and gives its inertia [2, 3, 7, 6].

For use in the order-ten proof, we state the retained matrix explicitly. Assume $b \ge 2a$. For $0 \le i, \ell < a$, set

$$n_i = b-a+i, \qquad N_{i\ell}=2(n_i-\ell)+1, \qquad R_\ell = 2b-2a+2+2\ell, \tag{9}$$

$$F_{i\ell}=\frac{1}{(2\ell+1)!}\prod_{r=0}^{2\ell-1}(2n_i-r)(2n_i+3+r), \tag{10}$$

with the empty product equal to one. Define

$$W_{i\ell}(z)=(z+2\ell+1)(2n_i+2\ell+z+1)^{\underline{2i}}, \qquad d_{i\ell h}=\frac{1}{h!}\sum_{j=0}^{h}(-1)^{h-j}\binom{h}{j}W_{i\ell}(N_{i\ell}-j). \tag{11}$$

The $d_{i\ell h}$ are integers because they are Newton coefficients of an integer-valued polynomial. With independent symbols $U_1,\dots,U_{2a-1}$, put

$$\mathcal{A}^{[a]}_{i\ell}=2F_{i\ell}+\frac{R_\ell!}{(2n_i+1)(2n_i+2)(2\ell+2)!N_{i\ell}!}\sum_{\substack{1\le h\le 2i+1\\ h\ \text{odd}}} d_{i\ell h}N_{i\ell}^{\underline{h}}U_{a-i+\ell+(h-1)/2}. \tag{12}$$

At the Bernoulli specialization

$$U_j=2B_{2b-2j}, \qquad 1\le j\le 2a-1, \tag{13}$$

the retained determinant controls the original source matrix.

Lemma 2.2 (Exact determinant multiplier; inherited)**.** *Let*

$$h_n=\frac{((2n+1)!)^2}{(4n+2)!(4n+3)!}, \qquad c_n=\frac{(2n+1)(2n+2)!}{(4n+2)!}, \qquad \epsilon_{a,b}=a(b-a)+\frac{a(a-1)}{2}.$$

For $b \ge 2a$,

$$\det M_{a,b}=(-1)^{\epsilon_{a,b}}\frac{\prod_{t=0}^{b-1}(2b+2t-2a+1)!}{\prod_{k=0}^{b-1}(2b-2k)!}\left(\prod_{n=0}^{b-a-1}h_n\right)\left(\prod_{i=0}^{a-1}c_{b-a+i}\right)\det\mathcal{A}^{[a]}. \tag{14}$$

Every factor outside $\det\mathcal{A}^{[a]}$ *is nonzero. Therefore*

$$\det M_{a,b}\neq 0 \quad\Longleftrightarrow\quad \det\mathcal{A}^{[a]}\neq 0.$$

The delivered verifier checks (14) modulo the certificate prime at $b=10,11,12,13,14$, comparing the direct source determinant with the retained determinant times the full multiplier. These spot checks are not needed once Theorem 2.2 is accepted, but they guard against normalization errors in the implementation.

2.3 Eventual nonsingularity

The retained determinant is a polynomial of total Bernoulli degree at most a after clearing explicit denominators. Its top homogeneous part is a positive Hankel moment determinant after Euler's formula for B_{2m} is substituted. The lower Bernoulli degrees are bounded by an explicit coefficient norm. Comparing the two produces a terminating threshold algorithm for each fixed a.

Theorem 2.3 (Effective eventual rank; inherited)**.** *For every fixed $a \geq 1$, there is a constructible B_a such that $\det M_{a,b} \neq 0$ for all $b \geq B_a$. The exact source computation gives*

$$B_1 = 16, \quad B_2 = 45, \quad B_3 = 142, \quad B_4 = 467, \quad B_5 = 1920.$$

In particular,

$$\det M_{5,b} \neq 0 \qquad (b \geq 1920). \tag{15}$$

The role of the present paper is therefore finite but decisive: prove nonsingularity for every $6 \leq b < 1920$ without leaving exceptional progressions.

3 Complete optimal-prefix reconstruction at smaller order ten

We now close the finite range left by Theorem 2.3. The argument is an exact certificate over one prime field. Its force comes from the analytic reduction already proved in the source: computation is used only after the problem has become a finite list of explicitly specified rational determinants.

3.1 A good prime and exact Bernoulli reduction

Fix

$$P = 1{,}000{,}003. \tag{16}$$

Trial division by the odd integers through 1000 proves that P is prime; this finite check is included in both verifiers. The Bernoulli numbers are reduced modulo P from their defining recurrence, not from a table. With $B_1 = -1/2$ and $m \geq 1$,

$$B_{2m} = \frac{\dfrac{2m-1}{2} - \displaystyle\sum_{j=1}^{m-1} \binom{2m+1}{2j} B_{2j}}{2m+1}. \tag{17}$$

This follows immediately from $\sum_{j=0}^{2m} \binom{2m+1}{j} B_j = 0$ and the vanishing of odd Bernoulli numbers beyond B_1.

For the retained matrices in the range $10 \leq b \leq 1919$, the largest factorial argument occurring in a matrix entry is 3838, the largest denominator appearing in the Bernoulli recurrence is 3839, and the largest Bernoulli number used by an entry is B_{3836}. The scalar multiplier in (14) contains factorials only up to 7674. All these integers are strictly below P. It follows inductively from (17) that every rational Bernoulli number occurring in the certificate has denominator prime to P. Every factorial and every remaining explicit linear denominator is likewise a unit in $\mathbb{F}_P$. Thus all source formulas admit an unambiguous reduction modulo P.

Lemma 3.1 (Denominator-safe reduction)**.** *For $6 \leq b \leq 9$, reduction of the source matrix $M_{5,b}$ modulo P is defined entrywise. For $10 \leq b \leq 1919$, reduction of the retained matrix $\mathcal{A}^{[5]}$ and of the multiplier in (14) is defined, and the multiplier is nonzero in $\mathbb{F}_P$. Consequently,*

$$\det \mathcal{A}^{[5]} \not\equiv 0 \pmod{P} \quad \Longrightarrow \quad \det M_{5,b} \neq 0 \quad \text{in } \mathbb{Q}.$$

The analogous implication with the direct determinant holds for $6 \le b \le 9$.

Proof. For the four direct cases the largest factorial argument is 25. In the retained range, inspection of (9)– (12) gives the bounds stated above. Every factor in (14) is a product or quotient of nonzero integers less than P, so it is nonzero in $\mathbb{F}_P$. Reduction modulo P is therefore a ring homomorphism on the localization of $\mathbb{Z}$ generated by all denominators in question. A zero rational determinant would reduce to zero; the contrapositive proves the claim. □

3.2 The exhaustive residue vector

Define the proof residue $r_b \in \mathbb{F}_P$ by

$$r_b = \begin{cases} \det M_{5,b} \pmod P, & 6 \le b \le 9, \\ \det \mathcal{A}^{[5]}(b; 2B_{2b-2}, \dots, 2B_{2b-18}) \pmod P, & 10 \le b \le 1919. \end{cases} \tag{18}$$

The split at $b = 10 = 2a$ is important: below it, the generic retained formula has endpoint terms not covered by its derivation, so the source matrix is used directly.

Theorem 3.2 (One-prime finite completion)**.** *For every integer* b *with* $6 \le b \le 1919$,

$$r_b \neq 0 \quad \text{in } \mathbb{F}_{1,000,003}. \tag{19}$$

Hence $\det M_{5,b} \neq 0$ *for every such* b.

Proof. The exact program `generate_order10_certificate.py` performs the following finite calculation.

1. It constructs factorials and inverse factorials through 3839 in $\mathbb{F}_P$ and generates all required Bernoulli residues from (17).
2. For $6 \le b \le 9$ it builds (7); for $10 \le b \le 1919$ it builds the five-square matrix (12) at (13).
3. It computes each determinant by exact Gaussian elimination in $\mathbb{F}_P$ and records all 1914 residues.

Every recorded residue lies in $\{1, \dots, P-1\}$. The complete vector is contained in `verification/order10_modular_certificate.json`. Its canonical comma-separated encoding has SHA-256 digest

```
3977e54f5b43db43e8f080c32f2e5b22
        beac54412cca32ff5d03d84975111836.
```
(20)

For orientation, five entries are

b	6	8	10	962	1919
r_b	484126	572800	668726	989617	667530

The value at $b = 962$ lies in the exceptional 2-adic progression left open by the preceding order-ten report; it is nonzero here.

A second verifier replays the entire calculation without importing the first implementation. It uses the full recurrence $\sum_{k=0}^{n} \binom{n+1}{k} B_k = 0$ for every Bernoulli number and a row-by-row subset

expansion for determinants, rather than the even-only recurrence and Gaussian elimination. It reproduces all 1914 residues and the digest (20). The digest is an integrity check, not a substitute for the supplied vector and replay programs. Finally, Theorem 3.1 converts each modular nonvanishing statement into rational nonvanishing. □

The implementation also checks the complete multiplier (14) at $b = 10, 11, 12, 13, 14$ by comparing the direct determinant with the retained determinant times every scalar factor. These checks defend against an implementation normalization error; the mathematical justification for all $b \geq 10$ is the inherited exact multiplier lemma.

3.3 The completed fixed-order theorem

Theorem 3.3 (Complete smaller-order-ten reconstruction)**.** *For every integer $b \geq 6$,*

$$\boxed{\det M_{5,b} \neq 0.} \tag{21}$$

Equivalently, for every even $q > 10$, the first q centered moments for the harmonic-tail pair $(p, q) = (10, q)$ have full formal rank q on the specified coordinate list

$$C_0, \ldots, C_{q/2-1}, \quad \zeta(0), \zeta(2), \ldots, \zeta(q-2).$$

Proof. The finite interval $6 \leq b \leq 1919$ is Theorem 3.2. The half-line $b \geq 1920$ is (15). The rank conclusion follows from Theorem 2.1. □

Remark 3.4 (What the theorem does and does not say)**.** The result completes one infinite fixed-order family and resolves the first explicit task in the research agenda of the quartic reconstruction report. It does not prove the unrestricted conjecture for every $a < b$, and it does not assert arithmetic independence of even zeta values. The proof is conventional exact mathematics with a finite computer-verified lemma; it has not been formalized in a proof assistant or independently refereed.

4 Ordinary convergent reconstruction and polygamma identities

The rank theorem is more than a statement about a determinant. It allows every late zeta coordinate to be extracted by a finite rational combination of centered moments. The analytic continuation can then be converted into an ordinary absolutely convergent series by subtracting only the nonnegative powers in the centered asymptotic expansion.

4.1 Centered asymptotic coefficients

For an integer $s \geq 2$ and $x = n + 1/2$, the shifted Euler–Maclaurin expansion [5] is

$$A_s(n) \sim \sum_{m \geq 0} \alpha_{s,m} x^{1-s-2m}, \tag{22}$$

where

$$\alpha_{s,0} = \frac{1}{s-1}, \qquad \alpha_{s,m} = \frac{B_{2m}(1/2)}{(2m)!}(s)_{2m-1} \quad (m \geq 1). \tag{23}$$

Here $(s)_j = s(s+1)\cdots(s+j-1)$ and $B_{2m}(1/2) = (2^{1-2m}-1)B_{2m}$. The half-integer center removes all odd corrections. For the order-ten pair define

$$\beta_{b,\ell} = \sum_{u=0}^{\ell} \alpha_{10,u}\alpha_{2b,\ell-u}. \tag{24}$$

Then

$$A_{10}(n)A_{2b}(n) \sim \sum_{\ell\ge 0} \beta_{b,\ell} x_n^{-2b-8-2\ell}. \tag{25}$$

4.2 The rational extraction rows

Let $N^{(b)} = M_{5,b}^{-1}$, which exists by Theorem 3.3. For $0 \le k < b$ and $0 \le r \le 2b-1$, set

$$w_{b,k,r} = \sum_{t=0}^{b-1} N_{kt}^{(b)} \binom{2(b+t)+1}{2r} 2^{-2(b+t-r)}, \tag{26}$$

where the binomial coefficient is understood to be zero when $r > b+t$. Define the even polynomial

$$P_{b,k}(x) = \sum_{r=0}^{2b-1} w_{b,k,r} x^{2r}. \tag{27}$$

Its nonnegative asymptotic part against the product of tails is

$$Q_{b,k}(x) = \sum_{r=0}^{2b-1} w_{b,k,r} \sum_{\substack{\ell\ge 0 \\ 2r-2b-8-2\ell\ge 0}} \beta_{b,\ell} x^{2r-2b-8-2\ell}. \tag{28}$$

Every sum is finite and rational. In particular,

$$\deg P_{b,k} \le 4b-2, \qquad \deg Q_{b,k} \le 2b-10. \tag{29}$$

Theorem 4.1 (All-coordinate ordinary reconstruction)**.** *For every $b \ge 6$ and $0 \le k < b$,*

$$\boxed{\sum_{n=0}^{\infty} \{P_{b,k}(x_n)A_{10}(n)A_{2b}(n) - Q_{b,k}(x_n)\} = \zeta(2k),} \qquad x_n = n + \frac{1}{2}, \tag{30}$$

where $\zeta(0) = -1/2$. The series is absolutely convergent, with summand $O(n^{-2})$.

Proof. From (6),

$$Z_k = \sum_{t=0}^{b-1} N_{kt}^{(b)} C_{b+t}.$$

Substitution of (4) gives

$$Z_k = \sum_{r=0}^{2b-1} w_{b,k,r} D_r.$$

By definition, the right side is the centered regularized sum of $P_{b,k}(x_n)A_{10}(n)A_{2b}(n)$. Equations (25) and (28) show that subtraction of $Q_{b,k}$ removes every nonnegative power, leaving $O(x_n^{-2})$. Hence the subtracted series converges absolutely.

It remains to compare the ordinary sum with the centered regularized one. Every monomial in $Q_{b,k}$ has nonnegative even degree, and

$$\sum_{n\geq 0}^{\text{a.c.}}(n+\tfrac{1}{2})^{2j}=\zeta(-2j,\tfrac{1}{2})=0 \qquad (j\geq 0).$$

Thus the finite subtraction has zero centered zeta-regularized sum. The ordinary subtracted series therefore equals the inherited regular value $\sum_r w_{b,k,r}D_r = Z_k = \zeta(2k)$. □

Corollary 4.2 (Product-polygamma form)**.** *For every $b \geq 6$ and $0 \leq k < b$,*

$$\sum_{n=0}^{\infty}\left\{\frac{P_{b,k}(x_n)}{9!(2b-1)!}\psi^{(9)}(n+1)\psi^{(2b-1)}(n+1)-Q_{b,k}(x_n)\right\}=\zeta(2k). \tag{31}$$

Proof. For even s, equation (2) reads $A_s(n) = \psi^{(s-1)}(n+1)/(s-1)!$. Substitute this into (30). □

Remark 4.3 (Effective, not merely existential)**.** All coefficients in (30) are obtained by finite rational arithmetic: form (7), invert it, apply (26), and truncate the explicit asymptotic convolution (28). The accompanying identity generator follows exactly this procedure. No recognition of numerical constants or fit to a floating-point series is involved.

5 A multiple-polylogarithm generator for two harmonic tails

The moment reconstruction has a natural analytic generating function. We use the nested-sum convention

$$\mathrm{Li}_{r,s}(x,y)=\sum_{m>n\geq 1}\frac{x^m y^n}{m^r n^s}, \tag{32}$$

initially in its domain of absolute convergence. This is a depth-two multiple polylogarithm in two arguments; the notation is chosen to make the larger and smaller summation indices visible [8, 9].

Theorem 5.1 (Two-tail polylogarithmic generator)**.** *Let $p, q > 1$ be integers and $|z| < 1$. Then*

$$\boxed{\mathcal{F}_{p,q}(z):=\sum_{n=0}^{\infty}A_p(n)A_q(n)z^n=\frac{\zeta(p)\zeta(q)-\mathrm{Li}_{p,q}(1,z)-\mathrm{Li}_{q,p}(1,z)-\mathrm{Li}_{p+q}(z)}{1-z}.} \tag{33}$$

The same function has the Gamma-normalized double Mellin integral

$$\mathcal{F}_{p,q}(z)=\frac{1}{\Gamma(p)\Gamma(q)}\int_0^\infty\int_0^\infty\frac{t^{p-1}u^{q-1}e^{-t-u}}{(1-e^{-t})(1-e^{-u})(1-ze^{-t-u})}\,\mathrm{d}t\,\mathrm{d}u. \tag{34}$$

Proof. Expand the two tails and interchange absolutely convergent sums:

$$\begin{aligned}\mathcal{F}_{p,q}(z)&=\sum_{m,k\geq 1}\frac{1}{m^p k^q}\sum_{0\leq n<\min(m,k)}z^n\\&=\frac{1}{1-z}\sum_{m,k\geq 1}\frac{1-z^{\min(m,k)}}{m^p k^q}.\end{aligned}$$

The part without $z^{\min(m,k)}$ is $\zeta(p)\zeta(q)$. Splitting the remaining double sum into $m > k$, $k > m$, and $m = k$ gives respectively $\mathrm{Li}_{p,q}(1,z)$, $\mathrm{Li}_{q,p}(1,z)$, and $\mathrm{Li}_{p+q}(z)$, proving (33).

For (34), use

$$A_p(n) = \frac{1}{\Gamma(p)} \int_0^\infty \frac{t^{p-1} e^{-(n+1)t}}{1 - e^{-t}} \, \mathrm{d}t$$

and the analogous formula for $A_q(n)$. Summing the resulting geometric series in ze^{-t-u} gives the stated integral. Positivity for real $z \in [0, 1)$ and absolute convergence for $|z| < 1$ justify the interchanges; the complex identity then follows by holomorphy. □

5.1 Centered weights as one differential operator

Let

$$\mathcal{E} = z\frac{\mathrm{d}}{\mathrm{d}z} + \frac{1}{2}. \tag{35}$$

Because $\mathcal{E} z^n = (n + 1/2)z^n$, every polynomial weight is encoded by functional calculus in this single operator:

$$P(\mathcal{E})\mathcal{F}_{p,q}(z) = \sum_{n=0}^{\infty} P(n + \tfrac{1}{2}) A_p(n) A_q(n) z^n. \tag{36}$$

Likewise,

$$Q(\mathcal{E})\frac{1}{1 - z} = \sum_{n=0}^{\infty} Q(n + \tfrac{1}{2}) z^n. \tag{37}$$

This gives a compact polylogarithmic normal form for every identity in Theorem 4.1.

Corollary 5.2 (Polylogarithmic boundary reconstruction)**.** *For $b \geq 6$ and $0 \leq k < b$,*

$$\boxed{\lim_{z \uparrow 1} \left\{ P_{b,k}(\mathcal{E})\mathcal{F}_{10,2b}(z) - Q_{b,k}(\mathcal{E})\frac{1}{1 - z} \right\} = \zeta(2k),} \tag{38}$$

where $\mathcal{F}_{10,2b}$ is given explicitly by (33). The limit may be taken radially in the unit disk.

Proof. Equations (36) and (37) show that the expression in braces has Taylor coefficients equal to the summand of (30). That coefficient series is absolutely summable, so Abel's theorem identifies the radial limit with its sum. □

Repeated argument differentiation introduces no new function class: for every integer $j \geq 0$,

$$\left(z\frac{\mathrm{d}}{\mathrm{d}z}\right)^j \mathrm{Li}_{r,s}(1, z) = \mathrm{Li}_{r,s-j}(1, z), \qquad \left(z\frac{\mathrm{d}}{\mathrm{d}z}\right)^j \mathrm{Li}_r(z) = \mathrm{Li}_{r-j}(z). \tag{39}$$

Thus (38) is a finite identity among ordinary polylogarithms, depth-two multiple polylogarithms with shifted orders, and rational functions. Formula (34) supplies a parallel Gamma-integral representation of the same boundary values.

Remark 5.3 (Why the numerator is a divided tail)**.** The factor $(1 - z)^{-1}$ in (33) is not an artifact of summation. It records the finite geometric prefix $1 + z + \cdots + z^{\min(m,k)-1}$. The numerator therefore measures exactly the difference between the unrestricted product $\zeta(p)\zeta(q)$ and the three regions in which the smaller index has already crossed the prefix boundary.

6 Integer-shift Stieltjes correlations from a regularized depth-two kernel

The tail generator admits a useful singular-order limit. Write the Laurent expansion of the Hurwitz zeta function at one as

$$\zeta(1+\alpha,a)=\frac{1}{\alpha}+\sum_{r=0}^{\infty}\frac{(-1)^r}{r!}\gamma_r(a)\alpha^r, \tag{40}$$

where $\gamma_r(1)=\gamma_r$ and $\gamma_0=\gamma$; see, for example, [5, 10]. At integer shifts,

$$\gamma_r(n+1)=\gamma_r-\sum_{m=1}^{n}\frac{\log^r m}{m}. \tag{41}$$

This follows by comparing (40) in $\zeta(1+\alpha,n+1)=\zeta(1+\alpha)-\sum_{m=1}^{n}m^{-1-\alpha}$.

6.1 Single-index order derivatives

Proposition 6.1 (Integer-shift Stieltjes generator)**.** *For every $r\geq 0$ and $|z|<1$,*

$$\boxed{\sum_{n=0}^{\infty}\gamma_r(n+1)z^n=\frac{\gamma_r-(-1)^r\,\partial_s^r\,\mathrm{Li}_s(z)|_{s=1}}{1-z}.} \tag{42}$$

Proof. Sum (41) over n and interchange the absolutely convergent sums. The logarithmic Dirichlet series is

$$\sum_{m\geq 1}\frac{z^m\log^r m}{m}=(-1)^r\,\partial_s^r\,\mathrm{Li}_s(z)|_{s=1}\,.$$

□

The point of the next theorem is that products do not require an uncontrolled double convolution of the shifted constants. A single pole-subtracted depth-two kernel generates all of them.

6.2 The bivariate regularized kernel

For $|z|<1$, define

$$\begin{aligned}\mathcal{K}(\alpha,\beta;z)=\;&\mathrm{Li}_{1+\alpha,1+\beta}(1,z)-\frac{\mathrm{Li}_{1+\beta}(z)}{\alpha}\\&+\mathrm{Li}_{1+\beta,1+\alpha}(1,z)-\frac{\mathrm{Li}_{1+\alpha}(z)}{\beta}+\mathrm{Li}_{2+\alpha+\beta}(z).\end{aligned} \tag{43}$$

Although the displayed terms have poles, their specified differences are regular.

Lemma 6.2 (Holomorphy of the pole subtraction)**.** *For every fixed $|z|<1$, $\mathcal{K}(\alpha,\beta;z)$ extends holomorphically to a neighborhood of $(0,0)$.*

Proof. For $\Re\alpha>0$,

$$\mathrm{Li}_{1+\alpha,1+\beta}(1,z)=\sum_{n\geq 1}\frac{z^n}{n^{1+\beta}}\zeta(1+\alpha,n+1),$$

$$\mathrm{Li}_{1+\alpha,1+\beta}(1,z)-\frac{\mathrm{Li}_{1+\beta}(z)}{\alpha}=\sum_{n\geq 1}\frac{z^n}{n^{1+\beta}}\left(\zeta(1+\alpha,n+1)-\frac{1}{\alpha}\right).$$

The parenthesis is holomorphic at $\alpha=0$. On compact subsets of a small bidisk it grows at most polynomially in $\log(n+1)$, whereas $|z|^n$ decays exponentially. The series is therefore locally normally convergent and holomorphic in both variables. The second pole-subtracted term is symmetric, and the ordinary polylogarithm in (43) is already holomorphic in its order near two. □

Put

$$G(\alpha)=\zeta(1+\alpha)-\frac{1}{\alpha}=\sum_{r\geq 0}\frac{(-1)^r\gamma_r}{r!}\alpha^r. \tag{44}$$

Theorem 6.3 (All-index Stieltjes correlation generator)**.** *For $|z|<1$ and (α,β) near $(0,0)$,*

$$\boxed{\sum_{n=0}^{\infty}\left(\zeta(1+\alpha,n+1)-\frac{1}{\alpha}\right)\left(\zeta(1+\beta,n+1)-\frac{1}{\beta}\right)z^n=\frac{G(\alpha)G(\beta)-\mathcal{K}(\alpha,\beta;z)}{1-z}.} \tag{45}$$

Consequently, for all $r,s\geq 0$,

$$\boxed{\sum_{n=0}^{\infty}\gamma_r(n+1)\gamma_s(n+1)z^n=\frac{\gamma_r\gamma_s-(-1)^{r+s}r!s!\,[\alpha^r\beta^s]\mathcal{K}(\alpha,\beta;z)}{1-z}.} \tag{46}$$

Proof. Apply Theorem 5.1 formally first in the half-plane $\Re\alpha,\Re\beta>0$, with $p=1+\alpha$ and $q=1+\beta$. Also use the one-tail identity

$$\sum_{n\geq 0}\zeta(1+\alpha,n+1)z^n=\frac{\zeta(1+\alpha)-\mathrm{Li}_{1+\alpha}(z)}{1-z}.$$

Subtract $1/\alpha$ from the first factor and $1/\beta$ from the second. The zeta terms combine into $G(\alpha)G(\beta)$, while the remaining polylogarithmic terms combine into $-\mathcal{K}$. This proves (45) in a nonempty domain. The holomorphic extensions from Theorem 6.2 continue it to a neighborhood of the origin. Coefficient extraction using (40) gives (46). □

6.3 Digamma square and an anchored antiderivative

Since $\gamma_0(n+1)=\gamma-H_n=-\psi(n+1)$, the lowest coefficient of Theorem 6.3 simplifies completely.

Corollary 6.4 (Digamma-square generating function)**.** *For $|z|<1$,*

$$\boxed{\sum_{n=0}^{\infty}(\gamma-H_n)^2z^n=\sum_{n=0}^{\infty}\psi(n+1)^2z^n=\frac{(\gamma+\log(1-z))^2+\mathrm{Li}_2(z)}{1-z}.} \tag{47}$$

Proof. The elementary generating functions

$$\sum_{n\geq 0}H_nz^n=-\frac{\log(1-z)}{1-z},\qquad\sum_{n\geq 0}H_n^2z^n=\frac{\mathrm{Li}_2(z)+\log^2(1-z)}{1-z}$$

follow by summing the defining single and double harmonic sums. Expand $(\gamma-H_n)^2$ and combine them. □

Termwise integration provides the anchored primitive requested by the factor $1/(n+1)$.

Theorem 6.5 (Anchored digamma-square primitive)**.** *For* $0 \le z < 1$,

$$\sum_{n=0}^{\infty} \frac{(\gamma - H_n)^2}{n+1} z^{n+1} = \frac{\gamma^3 - (\gamma + \log(1-z))^3}{3} - \operatorname{Li}_2(z)\log(1-z) - I(z), \tag{48}$$

$$\begin{aligned} I(z) &:= \int_0^z \frac{\log^2(1-t)}{t}\,\mathrm{d}t \\ &= 2\zeta(3) + \log^2(1-z)\log z + 2\log(1-z)\operatorname{Li}_2(1-z) - 2\operatorname{Li}_3(1-z). \end{aligned} \tag{49}$$

In particular,

$$\boxed{\sum_{n=0}^{\infty} \frac{(\gamma - H_n)^2}{(n+1)2^{n+1}} = \frac{\gamma^3 - (\gamma - \log 2)^3}{3} + \frac{\pi^2 \log 2}{12} - \frac{\log^3 2}{6} - \frac{\zeta(3)}{4}.} \tag{50}$$

Proof. Integrate (47) from 0 to z. The square term is integrated by the substitution $u = \gamma + \log(1-t)$. Integration by parts gives

$$\int_0^z \frac{\operatorname{Li}_2(t)}{1-t}\,\mathrm{d}t = -\operatorname{Li}_2(z)\log(1-z) - I(z).$$

Differentiating the right side of (49) and checking its zero value at $z = 0$ proves the closed form. At $z = 1/2$, substitute the classical evaluations

$$\operatorname{Li}_2(\tfrac{1}{2}) = \frac{\pi^2}{12} - \frac{\log^2 2}{2}, \qquad \operatorname{Li}_3(\tfrac{1}{2}) = \frac{7}{8}\zeta(3) - \frac{\pi^2 \log 2}{12} + \frac{\log^3 2}{6}$$

and simplify. □

More generally, integrating (46) gives an anchored antiderivative for every pair of indices:

$$\sum_{n=0}^{\infty} \frac{\gamma_r(n+1)\gamma_s(n+1)}{n+1} z^{n+1} = \int_0^z \frac{\gamma_r\gamma_s - (-1)^{r+s} r!\,s!\,[\alpha^r\beta^s]\mathcal{K}(\alpha,\beta;t)}{1-t}\,\mathrm{d}t. \tag{51}$$

The point is not merely the integral notation: the integrand is a finite coefficient of the explicitly specified depth-two kernel (43), so order derivatives and anchored primitives are controlled by one common analytic object.

7 An explicit $(10, 16)$ reconstruction of $\zeta(2)$

We record one fully expanded member of Theorem 4.1. The pair $(10, 16)$ is the first gap-six instance beyond the adjacent and gap-four order-ten cases. Let

$$L = 14{,}612{,}936{,}162{,}752{,}272{,}720{,}199{,}680 \tag{52}$$

and define

$$\mathscr{P}(x) = \sum_{r=0}^{15} c_r x^{2r}, \tag{53}$$

where the integer coefficients c_r are listed in Table 1. Also put

$$\mathscr{R}(x) = 1{,}552{,}076{,}939{,}623{,}430{,}612{,}844{,}544$$

$$
\begin{aligned}
&+ 2{,}244{,}965{,}091{,}528{,}662{,}071{,}640{,}064\,x^2 \\
&+ 169{,}465{,}732{,}042{,}429{,}783{,}080{,}960\,x^4 \\
&+ 2{,}907{,}927{,}622{,}440{,}764{,}571{,}648\,x^6.
\end{aligned} \tag{54}
$$

Theorem 7.1 (Explicit ordinary identity)**.** *With* $x_n = n + 1/2$*,*

$$
\boxed{\sum_{n=0}^{\infty} \Big\{ 135\,\mathscr{P}(x_n)[\zeta(10) - H_n^{(10)}][\zeta(16) - H_n^{(16)}] + \mathscr{R}(x_n) \Big\} = L\zeta(2).} \tag{55}
$$

The summand is $O(n^{-2})$*. Equivalently,*

$$
\sum_{n=0}^{\infty} \left\{ \frac{135\,\mathscr{P}(x_n)}{9!15!} \psi^{(9)}(n+1)\psi^{(15)}(n+1) + \mathscr{R}(x_n) \right\} = L\zeta(2). \tag{56}
$$

Proof. Take $b = 8$ and $k = 1$ in Theorem 4.1. Exact inversion of the rational matrix $M_{5,8}$ gives

$$
P_{8,1}(x) = \frac{\mathscr{P}(x)}{108{,}243{,}971{,}575{,}942{,}760{,}890{,}368}.
$$

The nonnegative asymptotic part (28) is

$$
\begin{aligned}
Q_{8,1}(x) = &-\frac{23127748662582496}{217749717276577245} - \frac{1238984692719613}{8064804343576935}x^2 \\
&- \frac{168349079333176}{14516647818438483}x^4 - \frac{14443832747344}{72583239092192415}x^6.
\end{aligned}
$$

Multiplying $\sum_n (P_{8,1}A_{10}A_{16} - Q_{8,1}) = \zeta(2)$ by L, noting that L is 135 times the denominator of $P_{8,1}$, gives (55). Equation (56) follows from (2). The exact generator and verifier reconstruct all displayed integers from (7), rather than checking a pre-entered formula. □

The high-precision script evaluates the ordinary series directly and agrees with $\zeta(2)$ after normalization to more than forty decimal places. This is recorded only as a diagnostic; the proof is the exact matrix inversion and asymptotic subtraction above.

8 Audit findings, corrections, and integration consequences

The source formulas used in the proof were rederived at the interfaces where normalization mistakes are most likely: the omitted-binomial convention in $M_{a,b}$, the factor two in the Bernoulli coordinates $U_j = 2B_{2b-2j}$, the retained-column reversal, and the scalar multiplier in (14). No mathematical contradiction was found in those inherited formulas. The following status and presentation corrections should nevertheless be made when this package is integrated.

8.1 Research-status updates

A1. The statement that the smaller-order-ten family remains open is now obsolete. It should be replaced by Theorem 3.3: every $M_{5,b}$ with $b \geq 6$ is nonsingular.

A2. The exceptional congruence class $b \equiv 962 \pmod{2048}$ in the preceding 2-adic theorem was a limitation of that congruence, not a candidate zero set. The new finite certificate includes every member of this class below 1920, beginning with $b = 962$, and the inherited eventual theorem covers all members at or above 1920.

A3. The broad unequal-order conjecture is still open. Completion of $p = 10$ must not be described as a proof for all $p < q$. The next fixed smaller order is $p = 12$, and the all-order problem requires control uniform in a.

8.2 Certificate wording

The residue vector is deliberately mixed: for $b < 10$ it contains the direct source determinant, while for $b \geq 10$ it contains the normalized five-square determinant. Calling every entry "the residue of $\det M_{5,b}$" would be inaccurate. The exact definition is (18); equivalence to source nonvanishing follows from the nonzero multiplier. The JSON metadata and integration notes preserve this distinction.

A checksum alone is not a certificate. The proof objects are the full residue vector, the formulas generating each matrix, the denominator argument, and the replay programs. The SHA-256 value only detects accidental or silent modification of that data.

8.3 Analytic normalization

The ordinary identities require the centered coordinate $x_n = n + 1/2$. At this center the subtraction polynomial contains only even powers and every such monomial has zero Hurwitz-zeta regularized sum because $\zeta(-2j, 1/2) = 0$. Replacing x_n by $n+1$ without recomputing the asymptotic polynomial would introduce nonzero Bernoulli counterterms and invalidate (30). Likewise, omitting the constant term of $Q_{b,k}$ is not harmless: $\zeta(0, 1/2) = 0$ is exactly what permits that constant to be subtracted without changing the regularized value.

8.4 Novelty and literature scope

The inspected project already contains extensive work on harmonic-tail reconstruction, Stieltjes correlations, and multiple polylogarithms. This report claims the following relative contributions only:

- the exact completion of the $p = 10$ fixed-order family;
- the explicit all-coordinate ordinary identities that follow from that completion;
- the two-tail generator (33) in the normalization used here; and
- the pole-subtracted kernel (43) and its coefficient formula for integer-shift Stieltjes products.

A targeted literature search found the standard multiple-polylogarithm, Hurwitz-zeta, Bernoulli, and Stieltjes ingredients, but not these exact combined formulas in this normalization. This is not a comprehensive priority search, and no global first-discovery claim is made. The article remains an unrefereed research draft.

8.5 Recommended manuscript placement

The determinant theorem belongs immediately after the current fixed-order and eventual-rank material. The ordinary reconstruction and polylogarithmic generator can follow the harmonic-tail chapter as a new all-coordinate consequence. The Stieltjes kernel is logically independent of the order-ten certificate and may instead be placed in the shifted-Stieltjes chapter, with a cross-reference to the tail generator from which it is obtained by singular-order regularization. The supplied

integration fragment and claim ledger use namespaced labels to avoid collisions during that editorial split.

9 Further research questions and exact next targets

The results above suggest several concrete projects. Each target is phrased so that a successful answer has a checkable mathematical endpoint rather than a larger table of experiments.

R1. Complete smaller order twelve. Apply the fixed-size reduction with $a = 6$. First compute a rigorous eventual threshold and then certify the remaining finite interval with one or several denominator-safe primes. A stronger result would replace the finite scan by a structural congruence or a sign argument for the six-square determinant.

R2. Remove computation from the order-ten theorem. Find a recurrence, total-positivity statement, or arithmetic invariant showing $\det \mathcal{A}^{[5]} \neq 0$ for every $b \geq 10$. The modular vector in this package is useful training data, but the desired endpoint is a uniform proof explaining why cancellation cannot occur.

R3. Prove the unrestricted unequal-order conjecture. The eventual theorem controls b for fixed a, but its threshold grows with a. Seek a bound on the lower Bernoulli-degree terms that is uniform in the separation $b - a$, or a determinant factorization whose nonvanishing is manifest for all $1 \leq a < b$.

R4. Closed inverse rows. The polynomials $P_{b,k}$ are explicit through matrix inversion but lack a product formula. Derive closed coefficient rows, perhaps through the Padé orthogonal polynomials and Christoffel–Darboux kernels. A useful answer should specify every coefficient and denominator directly, not refer to an unevaluated inverse matrix.

R5. Minimal subtraction and accelerated identities. The present $Q_{b,k}$ leaves an $O(n^{-2})$ summand. Subtract additional negative asymptotic powers and compensate them by the known values $\zeta(2j, 1/2)$ to obtain arbitrarily rapid convergence. Determine a canonical acceleration with small rational coefficients and quantify its arithmetic denominator growth.

R6. Boundary values of the two-tail generator. Continue (33) through $z = 1$ after the complete Euler-operator subtraction. Express the finite boundary values in a minimal basis of multiple zeta values and order derivatives. The operator formula makes every candidate finite, but functional reduction at high shifted order remains nontrivial.

R7. Rationally shifted and character-twisted tails. Replace $A_p(n)$ by $\zeta(p, n+a)$ with rational a, or by a Dirichlet character tail. Determine the analogue of the three-region numerator in (33), the correct centered subtraction, and the resulting coordinate space of Dirichlet L-values.

R8. Higher products. For three or more tails, partition the summation region by the minimum index and derive depth-d multiple-polylogarithm generators. Then ask for finite-rank reconstruction theorems and ordinary convergent polygamma-product identities. The main challenge is to organize the ordered set partitions without hiding endpoint terms.

R9. Boundary Stieltjes correlations. Study the Abelian and finite-part limits of (46) as $z \to 1$. Identify exactly which combinations of Stieltjes constants, multiple-zeta derivatives, and powers of $\log(1 - z)$ survive. The pole-subtracted kernel should be continued jointly in (α, β, z) rather than coefficient by coefficient.

R10. Functional equations for the kernel. Investigate transformations $z \mapsto 1-z$, $z \mapsto z/(z-1)$, and roots of unity for $\mathcal{K}(\alpha,\beta;z)$. A complete formula must retain the order-derivative terms created by the pole subtractions. Special values could yield new exact sums of $\gamma_r(n+1)\gamma_s(n+1)$ with algebraic weights.

R11. Higher anchored primitives and multiple Gamma functions. Integrate the all-index kernel repeatedly and determine when the result reduces to classical polylogarithms, Barnes multiple Gamma functions, or genuinely new depth. The $r=s=0$ formula (48) is the first case of this hierarchy.

R12. Formal verification. Formalize the finite determinant reduction, the denominator-safe modular certificate, and the min-index proof of (33). The finite-field portion is well suited to reflection. A later layer can formalize the analytic continuation and centered regularization used in Theorem 4.1.

R13. Independent literature and priority audit. Search multiple-zeta, Euler-sum, and Stieltjes-constant literature for forms equivalent to (43) under stuffle or iterated-integral transformations. Record equivalence maps explicitly. This should be completed before any publication-level novelty claim is made.

A Certificate specification

This appendix records the finite proof object independently of any particular programming language.

A.1 Input range and residue meaning

The fixed parameters are

$$a = 5, \qquad P = 1{,}000{,}003, \qquad 6 \le b \le 1919.$$

For $b = 6,7,8,9$, the proof residue is the determinant of the direct $b\times b$ matrix (7). For $10 \le b \le 1919$, it is the determinant of the 5×5 matrix (12) after $U_j = 2B_{2b-2j}$. All operations are in $\mathbb{F}_P$.

A.2 Canonical generation algorithm

Listing 1: Mathematical pseudocode for the residue vector.

```
P = 1000003
build factorials and inverse factorials modulo P through 3839
B[0] = 1
for m = 1,...,1919:
    B[2m] = ((2m-1)/2
              - sum_{j=1}^{m-1} binom(2m+1,2j) B[2j])/(2m+1)

for b = 6,...,1919:
    if b < 10:
        R[b] = det(M_{5,b}) mod P
    else:
        substitute U_j = 2 B[2b-2j], j=1,...,9
        R[b] = det(A^[5]) mod P
    assert R[b] != 0
```

```
serialize R[6],R[7],...,R[1919] as comma-separated decimal integers
compute SHA256 of those bytes
```

The expected digest is (20). The JSON file also contains the matrix ranges, maximum indices, count, full vector, and sentinel entries. The certificate contains no random choices.

A.3 Two replay paths

The primary replay uses the even recurrence (17) and modular Gaussian elimination. The independent replay uses the full Bernoulli recurrence

$$B_n = -\frac{1}{n+1}\sum_{k=0}^{n-1}\binom{n+1}{k}B_k$$

directly for $0 \le n \le 3838$, verifies the vanishing of odd Bernoulli residues beyond B_1, and computes determinants by subset expansion. The two paths share only the mathematical formulas in the article and the published JSON result.

A.4 Sentinel values

b	matrix used	residue modulo P
6	direct $M_{5,6}$	484126
8	direct $M_{5,8}$	572800
10	retained $\mathcal{A}^{[5]}$	668726
962	retained $\mathcal{A}^{[5]}$	989617
1919	retained $\mathcal{A}^{[5]}$	667530

These values are convenient smoke tests, but validation requires the entire vector and digest.

B Coefficient table for the explicit identity

The coefficients in (53) are as follows. They are generated exactly by `verification/generate_explicit_identity.py`.

Table 1: Integer coefficients c_r in $\mathscr{P}(x) = \sum_{r=0}^{15} c_r x^{2r}$.

r	c_r
0	22571931032959635
1	18694493678772529244
2	2408312382491948637488
3	1118148189050534092814 72
4	2361587601371307705179904
5	2475790736948365983945 8304
6	1328291881310826646307 43040
7	3606053175177147649979 02336

r	c_r
8	45324743453528651478079897 6
9	15697617944859356742942720 0
10	-15193705758303175548652748 8
11	-14269627366498919165565337 6
12	-40996749143564725151334400
13	-47143359420364231928709 12
14	-20944973685099029594112 0
15	-2907927622440764571648

For a machine-readable record, `explicit_10_16_zeta2.json` stores the denominator of $P_{8,1}$, all sixteen numerators, and the four rational coefficients of $Q_{8,1}$.

C Reproducibility and evidential roles

All paths below are relative to the package root.

C.1 Exact proof checks

```
python3 verification/generate_order10_certificate.py
python3 verification/verify_order10_certificate.py
python3 verification/verify_order10_certificate_independent.py
python3 verification/verify_normalization_spots.py
python3 verification/generate_explicit_identity.py
python3 verification/verify_explicit_identity.py
```

The first two commands generate and replay the published modular vector. The third is the independently written full replay. The fourth checks the retained/source multiplier at five boundary values. The last two regenerate the explicit $(10, 16)$ identity from rational matrix inversion and centered asymptotics.

C.2 Numerical diagnostics

```
python3 verification/numerical_check_explicit_identity.py
python3 verification/numerical_check_generators.py
```

These scripts use high-precision numerical arithmetic to detect transcription or branch errors. They are not proof gates and are not interval-arithmetic certificates.

C.3 Complete replay and build

```
python3 verification/run_all.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The package has no compiled Python extension and no network dependency. The exact scripts use only the Python standard library; the numerical scripts require `mpmath`. The PDF build requires a conventional TEX Live installation with `latexmk`. The supplied `Makefile` exposes the targets `verify`, `pdf`, and `all`.

C.4 Claim classification

Class	Role
Inherited theorem	Shifted-Hankel reduction, exact determinant multiplier, and eventual threshold $B_5 = 1920$.
New exact theorem	Finite modular completion, complete order-ten rank, ordinary reconstruction, two-tail generator, and Stieltjes kernel.
Exact computation	Finite-field determinant vector and rational coefficient generation; both are replayable.
Numerical diagnostic	High-precision evaluation of displayed identities; useful for error detection but not invoked in proofs.
Open problem	Every item in Section 9; none is represented as settled.

References

[1] V. Reshetnikov and contributors, *ProveIt: Polylogarithms and their Arithmetic Bridges*, collective manuscript and incoming research, inspected at commit `04bf92d507588a26abadbaee1beb1e7350c8bffb` (11 October 2026). https://github.com/VladimirReshetnikov/ProveIt/tree/04bf92d507588a26abadbaee1beb1e7350c8bffb/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt research continuation, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, 11 October 2026, at the repository snapshot in [1]; especially the shifted-Hankel and general small-determinant sections.

[3] ProveIt research continuation, *Harmonic-Tail Reconstruction, Quartic Trigamma Moments, and Spectral Stieltjes Calculus*, 11 October 2026, at the repository snapshot in [1]; especially the complete order-eight theorem and the effective eventual-rank theorem.

[4] ProveIt research continuation, *Alternating Tails and Higher Spectral Defects*, 11 October 2026, at the repository snapshot in [1]; especially the five-square order-ten contraction and the higher-precision Bernoulli congruence.

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